MnPASS Express Lanes

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TH 169 Board Meeting

Your Destination...Our Priority

Overview

- Background & Context
- MnPASS Express Lanes
  - What they are
  - How they work
  - Why they're a key regional strategy
  - System vision
  - Use and performance
  - Corridor development
    - TH 169

Key Twin Cities Congestion Strategies

Highway System Management

- Regional Transportation Management Center (RTMC)
- Cameras & Loop Detectors on >90% of metro freeway system
- Traffic Information Options – internet, phone, radio/KBEM 88.5FM
- Dynamic Message Signs
- Ramp Meters (HOV/Bus Bypasses)
- FIRST Response Vehicles

Key Twin Cities Congestion Strategies cont.

Travel Demand Management – alternative modes (bike/ped), van/carpooling, park and rides, telework and other demand reduction strategies

Transit – LRT, BRT (hwy. & arterial), commuter rail, express bus and transit advantage strategies (bus only shoulders)

Land Use – policies encouraging greater transit and alternative mode travel, or a reduction in low occupancy vehicle trips on congested highways

Key Twin Cities Congestion Strategies cont.

Strategic Highway Investments –focus on low cost/high benefit hwy. projects and optimizing existing infrastructure & right-of-way

- CMSP – Congestion Mitigation & Safety Projects (ex. auxiliary lanes)
- Managed Lanes (ex. MnPASS Express Lanes)

Managed Lanes

Non-Priced Managed Lanes

- High Occupancy Vehicle Lanes T
- Smart Lanes
  - Intelligent Lane Control Signs
  - Variable Speed Limits
- Dynamic Shoulder Lanes

T=Transit Advantage

Managed Lanes

<u>Priced Managed Lanes = MnPASS Express Lanes</u>

- Typical MnPASS Lanes T
- Reversible MnPASS Lanes T
- Priced Dynamic Shoulder Lanes T

***Priced Managed Lanes are a form of congestion pricing and are also known as High Occupancy Toll (HOT) Lanes**

T=Transit Advantage

MnPASS Express Lanes

Current MnPASS Express Lanes
- I-394 – opened 2005
- I-35W – opened 2009

Operates during rush hour periods
- Buses, carpools (2 or more passengers), and motorcycles use for free
- Single occupant vehicles use for a fee (range: .25¢-$8.00) (average: $1.50)
- Open and free to all to use during non-peak rush hour periods

Utilizes electronic dynamic pricing
- Purpose of pricing is to maintain congestion free speeds and better utilize hwy. capacity

Typical MnPASS Express Lane

Why is MnPASS a key regional congestion strategy?

- Faster, safer more reliable travel options for people to choose from
- Critical to improving the regional bus transit system
- Proven to be a more efficient, cost-effective congestion management strategy – *High Return on Investment*
  - Maximizes hwy. performance and people throughput
  - Provides congestion-free reliability and sustainability that a traditional general purpose lane cannot
  - Reduces congestion in adjacent general purpose lanes
- Multimodal solution that can improve the state's economy, environment and quality of life

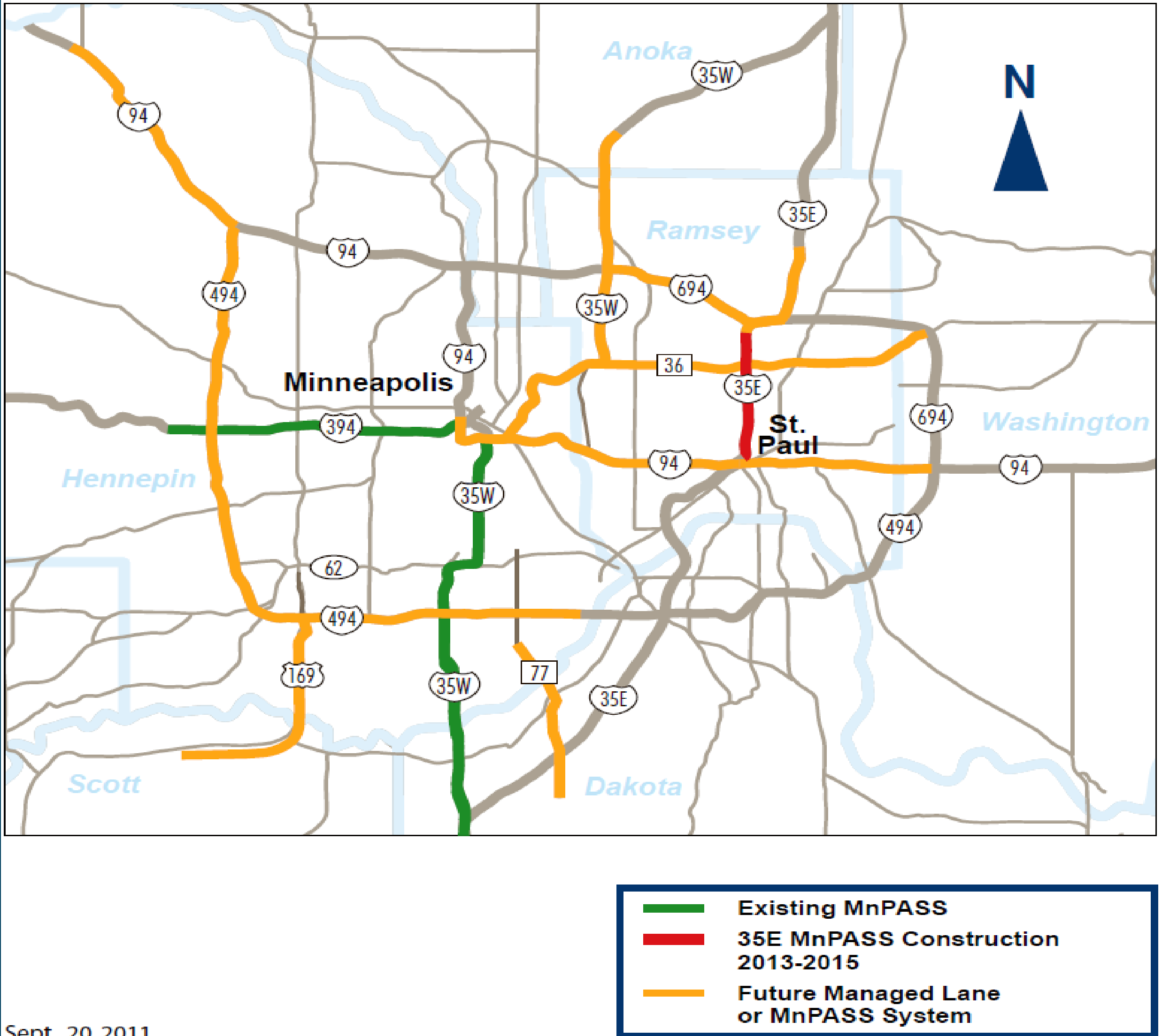
MnPASS System Study 2

Tier I
- I-35E north of St. Paul

Tier II
- I-35W north of Mpls.
- I-94 between Mpls. & St. Paul
- TH 36 (EB) between 35E & 35W

Tier III
- TH 77 (NB)
- I-494 (2 segments)
- TH 169

Regional 2030 Transportation Policy Plan
Future Managed Lane System



Sept. 20, 2011

MnPASS Use & Performance

- Steady annual growth (27,000 transponder holders)
  - I-35W MnPASS Express Lanes – 2012 trips up 24% over 2011 – active accounts up 22% – avg. lane speed 65mph
  - I-394 MnPASS Express Lanes – 2012 trips up 12% over 2011 – active accounts up 7% – avg. lane speed 59mph
- Greater than 80% satisfaction rate among customers
  - Time saved, congestion avoidance, choice and reliability valued most
  - Customers stay customers

MnPASS Use & Performance

- Transit users and operators strongly support system
  - Greater than 80% of the people using MnPASS are carpooling or riding transit
  - I-35W Express Bus Service – increased operating speeds (5-10 minute time savings); near perfect on-time performance; ridership up 27%; park and ride use up 20%; Lakeville Express fare box covers app. 70% of cost
- MnPASS lane can carry 50% more people than a general purpose lane during peak hour traffic
- Violation rates range from 5-15% depending on segment

MnPASS Revenues & Expenditures

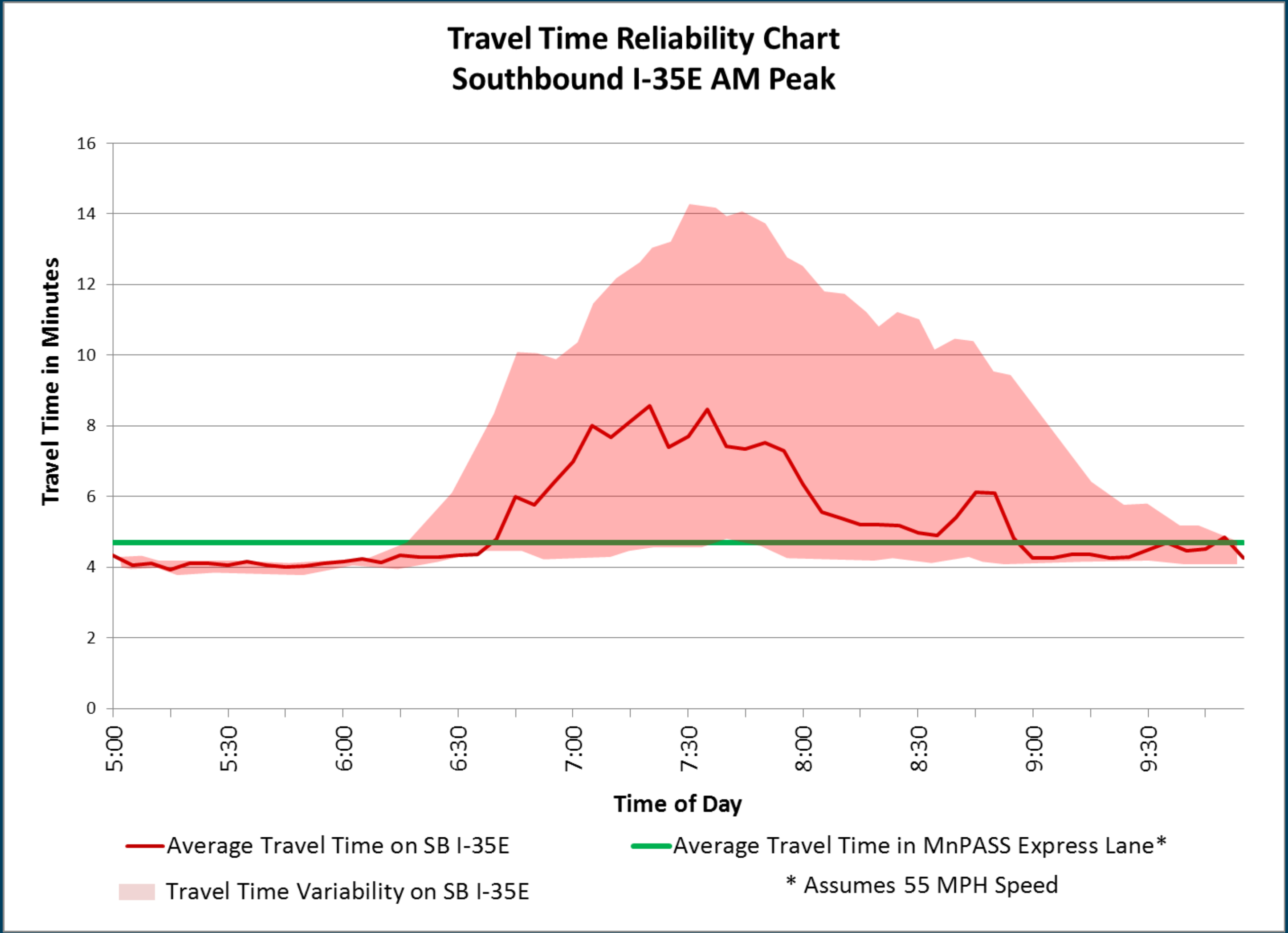
- Average toll: $1.32-$1.82
- Monthly transponder lease fee: $1.50

- 2012 Revenues & Expenditures
  - Revenue: Tolls-$2.5 million
Transponder Fees-$500,000
  - Operations & Maintenance Expenses: $2.4 million

- Purpose of pricing is to maximize traffic flow, *not* revenue

- Revenue use rules are unique to each corridor – generally revenue in excess of operations and maintenance costs on a corridor gets split 50/50 between MnDOT and the Met Council for hwy. and transit improvements in the corridor

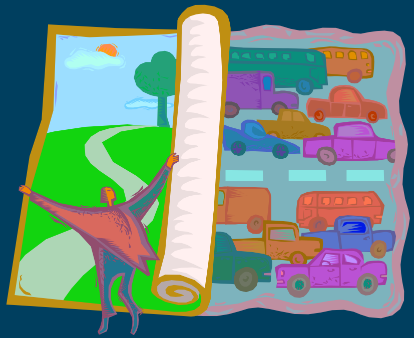
MnPASS Reliability



I-35W Urban Partnership Agreement Evaluation

- FHWA I-35W UPA Evaluation Report – nearing publication
  - Measured a variety of congestion and mobility factors before and after the I-35W UPA project which included the conversion of the I-35W HOV lane to a HOT lane and its extension into Mpls.
  - Key findings for the I-35W corridor included:
    - Peak hour travel time decreased 27% SB and 17% NB
    - Vehicle throughput increased 17%-49% depending on segment
    - Crash rates decreased by 20-25%
    - All corridor user groups benefitted from project
    - Benefit/Cost Analysis = 6.0
- I-35W improved even more in 2012

MnPASS Corridor Development



- I-35E MnPASS Project (St. Paul – Little Canada)
  - Beginning August 2013; Completion late 2015
- I-35W North Managed Lane Study (Mpls.- Forest Lake/Columbus)
  - Nearing completion
- I-94/TH 280 Managed Lane Study (Mpls.- St. Paul)
  - Underway – completion late 2013
- TH 77 (Bloomington – Apple Valley)
  - Underway – completion mid 2014
- I-35E MnPASS Extension Study (Little Canada-Vadnais Heights/White Bear Lake)
  - Beginning May 2013; Completion mid 2014
- I-494/I-94 Managed Lane Study (Airport-Albertville)
  - Beginning late 2013; Completion late 2014

Planning context for MnPASS and the TH 169 Corridor

- MnPASS System Study 2
  - TH 169 (I-494 – CR 17) – Tier 3
- Metro Hwy. System Investment Study
  - TH 169 (2 segments: MN River/TH 101-TH 62 and TH 62-I-394)
- 2030 Transportation Policy Plan
  - Managed lane vision est. cost $1.25B-$1.5B
  - Only $500M est. to be available for vision thru 2030
  - 2015-20 funding priority Tier 2 MnPASS corridors
  - 2020-30 funding priorities also include I-494 managed lane elements
- Highway Transitway/BRT Study
  - Includes TH 169 (I-394 thru Shakopee)

Thank You

- Questions, Comments, Suggestions

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Or visit

www.mnpass.org – General MnPASS website

http://www.dot.state.mn.us/cayugamnpass/index.html – I-35E MnPASS Project website